

Q1: To save energy for heating and cooling a building, an architect decides to design a part of the building in the basement in the shape of a rectangular cuboid. The total required floor area is $20,000 \text{ m}^2$, and dimensional constraints limit the length and width of the building to at most 50 m. According to aesthetic criteria, the ratio of length to width must equal the golden ratio 1.618. The height of each story must not exceed 3.5 m. The heating and cooling cost is 100 USD per square meter of exposed surface area of the building. The building owner requires that the annual energy cost of the building should not exceed 225,000 USD. Formulate the optimization problem to determine the optimal dimensions and number of stories that minimize the excavation cost.

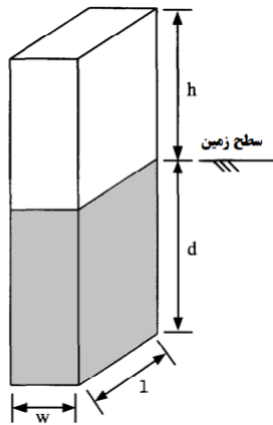


Figure 1: Building's schema and related parameters

Solution

According to the reference notes, we solve this problem in five steps.

Step 1: Clear Definition of the Problem

The objective is to design a partially underground building to minimize excavation costs while satisfying specific requirements for the total floor area, architectural dimensions (golden ratio), floor height, and annual energy costs.

Step 2: Data Collection

The data is given in the question itself.

Table 1: Design parameters and constraints for the basement building

Parameter	Value	Unit
Total required floor area	20,000	m^2
Maximum length / width	50	m
Golden ratio (length/width)	1.618	–
Maximum story height	3.5	m
Heating/cooling cost per exposed area	100	USD/ m^2
Maximum allowable annual energy cost	225,000	USD

Step 3: Definition of Design Variables

We must define the independent variables that describe the system and will be calculated during the design process.

- w : Width of the building (m)
- l : Length of the building (m)
- h : Height of the building above ground (m)
- d : Depth of the building underground (m)
- n : Total number of floors (integer)

Step 4: Identification of the Optimization Criterion

The optimization criterion must be defined as a function of the design variables to be minimized or maximized. The problem asks to minimize the excavation costs. Since excavation cost is directly proportional to the volume of the underground section, our objective function is to minimize this volume.

$$\text{Minimize } f(w, l, h, d, n) = l \cdot w \cdot d$$

Step 5: Identification and Modeling of Constraints

All design limitations and requirements must be defined in the form of equality or inequality constraints.

1. Total Floor Area (Equality): The total built area across all floors must be exactly 20,000 m.

$$n \cdot l \cdot w = 2000 \rightarrow h_1(X) : n \cdot l \cdot w - 20000 = 0$$

2. Golden Ratio (Equality): The length-to-width ratio must equal 1.618.

$$l/w = 1.618 \rightarrow h_2(X) : l - 1.618w = 0$$

3. Dimensional Limits (Inequality): Neither length nor width can exceed 50 m.

$$l \leq 50 \rightarrow g_1(X) : l - 50 \leq 0$$

$$w \leq 50 \rightarrow g_2(X) : w - 50 \leq 0$$

4. Floor Height Limit (Inequality): The average height of each floor (total height divided by the number of floors) cannot exceed 3.5 m.

$$\frac{h + d}{n} \leq 3.5 \rightarrow g_3(X) : h + d - 3.5n \leq 0$$

5. Energy Cost Limit (Inequality): The cost of heating and cooling is 100 per m² of the exposed area. The exposed area consists of the roof and the 4 above-ground walls. This cost must not exceed 225,000.

- Exposed Area = $(l \cdot w) + 2h(l + w)$
- Cost = $100 \cdot [l \cdot w + 2h(l + w)]$

$$100 \cdot [l \cdot w + 2h(l + w)] \leq 225000 \rightarrow g_4(X) : l \cdot w + 2h(l + w) - 2250 \leq 0$$

6. Physical Bounds (Inequality): Dimensions cannot be negative, and there must be at least one floor.

$$g_5(X) : -w < 0$$

$$g_6(X) : -l < 0$$

$$g_7(X) : -h < 0$$

$$g_8(X) : -d < 0$$

$$g_9(X) : 1 - n \leq 0$$

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

Minimize $f(X) = l \cdot w \cdot d$
 Subject to:

$$h_1(X) = n \cdot l \cdot w - 20000 = 0$$

$$h_2(X) = l - 1.618w = 0$$

$$g_1(X) = l - 50 \leq 0$$

$$g_2(X) = w - 50 \leq 0$$

$$g_3(X) = h + d - 3.5n \leq 0$$

$$g_4(X) = l \cdot w + 2h(l + w) - 2250 \leq 0$$

$$g_5(X) \text{ to } g_8(X) \leq 0 \text{ (non-negativity bounds)}$$

$$n \geq 1, \quad n \in \mathbb{Z}$$

Q2: To model the behavior of a unit in a power plant, several experiments have been carried out. The results of the inputs (X, Y) and output (Z) obtained from the experiments are presented in the table below. Based on prior knowledge of the system behavior, the most suitable mathematical model representing the system is the following relationship:

$$z = a_1x^2 + a_2y^2 + a_3xy$$

Formulate an optimal design problem to find the best coefficients a_1 , a_2 , and a_3 such that the given relationship has the greatest agreement with the experimental observations.

Table 2: Results of the experiments performed

Experiment No.	x	y	z
1	0	1	1.35
2	0.25	1	1.82
3	0.50	1	2.43
4	0.75	2	1.96
5	1	2	2.11
6	1.25	2	1.75
7	1.50	3	1.18
8	1.75	3	2.05
9	2	3	1.82

Solution

According to the reference notes, we solve this problem in five steps.

Step 1: Clear Definition of the Problem

Formulate an unconstrained optimization problem to find the optimal coefficients (a_1, a_2, a_3) for a mathematical model describing a power plant unit, such that the model best fits 9 sets of experimental data.

Step 2: Data Collection

The system's behavior is modeled by the equation:

$$z = a_1x^2 + a_2y^2 + a_3xy$$

We have $n = 9$ experimental data points (x_i, y_i, z_i) , placed in the table.

Step 3: Definition of Design Variables

Unlike physical design problems where dimensions are the variables, in data fitting, the coefficients of the model are the unknown variables we control.

- a_1 : Coefficient for x^2
- a_2 : Coefficient for y^2
- a_3 : Coefficient for xy

Step 4: Identification of the Optimization Criterion

To find the best fit, we must minimize the total error across all experiments. The standard, most robust metric for this is the Sum of Squared Errors (SSE).

For any given experiment i , the model's predicted output is:

$$\hat{z}_i = a_1 x_i^2 + a_2 y_i^2 + a_3 x_i y_i$$

The error for that experiment is the difference between the prediction and the actual measured value z_i :

$$e_i = \hat{z}_i - z_i = a_1 x_i^2 + a_2 y_i^2 + a_3 x_i y_i - z_i$$

The objective function $f(a_1, a_2, a_3)$ is the sum of the squares of these errors over all 9 experiments:

$$\text{Minimize } f(a_1, a_2, a_3) = \sum_{i=1}^9 e_i^2 = \sum_{i=1}^9 (a_1 x_i^2 + a_2 y_i^2 + a_3 x_i y_i - z_i)^2$$

Step 5: Identification and Modeling of Constraints

The problem statement does not provide any limits or required bounds for the coefficients a_1, a_2 , or a_3 . Therefore, this is an unconstrained optimization problem.

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

$\text{Minimize } f(a_1, a_2, a_3) = \sum_{i=1}^9 (a_1 x_i^2 + a_2 y_i^2 + a_3 x_i y_i - z_i)^2$ Subject to: No constraints.
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Q3: A manufacturer wants to use an alloy consisting of 50% tin, 30% lead, and 20% zinc. This alloy can be made by blending some of the available alloys whose compositions and costs are given in the table below. Formulate the optimal design problem for the best blend to make the required alloy with the least cost.

Table 3: Available alloys and their properties

Properties	A	B	C	D	E
Lead (%)	10	10	40	60	30
Zinc (%)	10	30	50	30	30
Tin (%)	80	60	10	10	40
Cost (\$/lb alloy)	8.2	9.3	11.2	13	17

Solution

According to the reference notes, we solve this problem in five steps.

Step 1: Clear Definition of the Problem

The objective is to formulate a mathematical model to find the minimum-cost mixture of five available alloys (A, B, C, D, E) that produces a new alloy containing exactly 50% Tin, 30% Lead, and 20% Zinc.

Step 2: Data Collection

From the problem description and the table, we extract the following data:

- Target Composition: 30% Lead (Pb), 20% Zinc (Zn), 50% Tin (Sn).
- Available Alloys which are defined in the table.

Step 3: Definition of Design Variables

The design variables represent the mass fractions of each available alloy used to create 1 pound of the target alloy.

- x_1 : Mass fraction of Alloy A
- x_2 : Mass fraction of Alloy B
- x_3 : Mass fraction of Alloy C
- x_4 : Mass fraction of Alloy D
- x_5 : Mass fraction of Alloy E

Step 4: Identification of the Optimization Criterion

The goal is to minimize the total cost of creating the alloy. The objective function is the sum of the costs of the individual fractions:

$$\text{Minimize } f(X) = 8.2x_1 + 9.3x_2 + 11.2x_3 + 13.0x_4 + 17.0x_5$$

Step 5: Identification and Modeling of Constraints

The constraints are the mass balance equations for each element to ensure the final alloy has the exact required percentages, plus the physical constraints that we cannot use negative amounts of an alloy.

1. Lead (Pb) Balance (Equality):

$$h_1(X) : 0.10x_1 + 0.10x_2 + 0.40x_3 + 0.60x_4 + 0.30x_5 = 0.30$$

2. Zinc (Zn) Balance (Equality):

$$h_2(X) : 0.10x_1 + 0.30x_2 + 0.50x_3 + 0.30x_4 + 0.30x_5 = 0.20$$

3. Tin (Sn) Balance (Equality):

$$h_3(X) : 0.80x_1 + 0.60x_2 + 0.10x_3 + 0.10x_4 + 0.40x_5 = 0.50$$

4. Total Mass Fraction (Equality):

The sum of all fractions must equal 1 (100%).

$$h_4(X) : x_1 + x_2 + x_3 + x_4 + x_5 = 1$$

5. Non-negativity Bounds (Inequality):

We cannot have a negative mass fraction of any alloy.

$$g_i(X) : -x_i \leq 0 \quad \text{for } i = 1, 2, 3, 4, 5$$

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

Minimize $f(X) = 8.2x_1 + 9.3x_2 + 11.2x_3 + 13x_4 + 17x_5$ Subject to: $h_1(X) = 0.1x_1 + 0.1x_2 + 0.4x_3 + 0.6x_4 + 0.3x_5 - 0.3 = 0$ $h_2(X) = 0.1x_1 + 0.3x_2 + 0.5x_3 + 0.3x_4 + 0.3x_5 - 0.2 = 0$ $h_3(X) = 0.8x_1 + 0.6x_2 + 0.1x_3 + 0.1x_4 + 0.4x_5 - 0.5 = 0$ $h_4(X) = x_1 + x_2 + x_3 + x_4 + x_5 - 1 = 0$ $g_1(X) = -x_1 \leq 0$ $g_2(X) = -x_2 \leq 0$ $g_3(X) = -x_3 \leq 0$ $g_4(X) = -x_4 \leq 0$ $g_5(X) = -x_5 \leq 0$
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Q4: A traveling salesman must visit n cities to sell his goods. He is required to start from city 1, visit $n - 1$ other distinct cities exactly once, and then return to city 1. Let d_{ij} be the distance between city i and city j . Formulate the problem as an optimization problem to minimize the total distance traveled by the salesman.

Solution

According to the reference notes, we solve this problem in five steps.

Step 1: Clear Definition of the Problem

Formulate a mathematical model for a traveling salesman to visit exactly n cities once, starting and ending at city 1, such that the total distance traveled is minimized.

Step 2: Data Collection

- Number of cities: n
- Starting/Ending node: City 1
- Distance matrix: d_{ij} (the distance from city i to city j)

Step 3: Definition of Design Variables

We need two types of variables to solve this problem effectively:

1. Binary Decision Variables (x_{ij}): These act as switches for the path.
 - $x_{ij} = 1$ if the salesman travels directly from city i to city j .
 - $x_{ij} = 0$ otherwise. (**Note:** $i, j \in \{1, 2, \dots, n\}$ and $i \neq j$ since you cannot travel from a city to itself)
2. Auxiliary Continuous Variables (u_i): These are required for the Miller-Tucker-Zemlin (MTZ) formulation to keep track of the sequence in which the cities are visited (for $i = 2, 3, \dots, n$). This prevents the creation of disconnected subtours.

Step 4: Identification of the Optimization Criterion

The goal is to minimize the total distance of the selected paths. We multiply the distance matrix by our binary decision variables and sum them up:

$$\text{Minimize } f(X) = \sum_{i=1}^n \sum_{j=1, j \neq i}^n d_{ij} x_{ij}$$

Step 5: Identification and Modeling of Constraints

The problem has several strict logical constraints to ensure a valid tour is formed:

1. Departure Constraint (Equality): The salesman must leave every city exactly once.

$$h_1(X) : \sum_{j=1, j \neq i}^n x_{ij} = 1 \quad \text{for all } i \in \{1, 2, \dots, n\}$$

2. Arrival Constraint (Equality): The salesman must arrive at every city exactly once.

$$h_2(X) : \sum_{i=1, i \neq j}^n x_{ij} = 1 \quad \text{for all } j \in \{1, 2, \dots, n\}$$

3. Subtour Elimination Constraint (Inequality): To ensure the path forms one single continuous loop involving all n cities and not multiple smaller loops, we apply the MTZ constraint to all cities except the starting city 1.

$$g_1(X) : u_i - u_j + nx_{ij} \leq n - 1 \quad \text{for } 2 \leq i \neq j \leq n$$

4. Variable Type Constraints:

$$x_{ij} \in \{0, 1\} \quad \text{for all } i, j$$

$$u_i \geq 0 \quad \text{for } i = 2, 3, \dots, n$$

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

Minimize $f(X) = \sum_{i=1}^n \sum_{j \neq i}^n d_{ij} x_{ij}$ Subject to: $\sum_{j \neq i}^n x_{ij} = 1 \quad \forall i \in \{1, \dots, n\}$ $\sum_{i \neq j}^n x_{ij} = 1 \quad \forall j \in \{1, \dots, n\}$ $u_i - u_j + nx_{ij} \leq n - 1 \quad \forall i, j \in \{2, \dots, n\}, i \neq j$ $x_{ij} \in \{0, 1\} \quad \forall i, j$ $u_i \geq 0 \quad \forall i \in \{2, \dots, n\}$

Q5: An optimal design problem is formulated as follows:

$$\text{Minimize } f(x_1, x_2) = x_1 x_2$$

subject to

$$\begin{aligned} x_1^2 + x_2^2 - 9 &\leq 0 \\ x_1 + x_2^2 &\leq 0 \end{aligned}$$

Find the optimal solutions graphically.

Solution

According to the reference notes, we solve this problem in six steps.

Step 1: Create a suitable coordinate system

Draw the x_1 - x_2 plane with x_1 on the horizontal axis and x_2 on the vertical axis.

Step 2: Draw the constraint boundaries

- g_1 : $x_1^2 + x_2^2 = 9$ is a circle of radius 3 centered at the origin.
- g_2 : $x_1 + x_2^2 = 0 \Rightarrow x_1 = -x_2^2$. This is a leftward-opening parabola with vertex at $(0, 0)$.

Step 3: Identify the feasible region for each constraint

- $x_1^2 + x_2^2 - 9 \leq 0$: interior and boundary of the circle.
- $x_1 + x_2^2 \leq 0$: region on or to the left of the parabola.

Step 4: Determine the overall feasible region

The feasible region is the intersection of the disk and the left-parabola region. It is symmetric about the x_1 -axis. The boundary consists of a circular arc for smaller $|x_2|$ and a parabolic arc for larger $|x_2|$, meeting at the intersection points of the circle and parabola. Solve

$$x_1^2 + x_2^2 = 9 \quad \& \quad x_1 = -x_2^2 \quad \Rightarrow \quad x_2^4 + x_2^2 - 9 = 0$$

Let $u = x_2^2$; then

$$u = \frac{-1 + \sqrt{37}}{2} \approx 2.541$$

The intersection points are

$$\left(\frac{1 - \sqrt{37}}{2}, \pm \sqrt{\frac{\sqrt{37} - 1}{2}} \right) \approx (-2.541, \pm 1.594)$$

Step 5: Draw the objective function contours

$f(x_1, x_2) = x_1 x_2 = C$ gives hyperbolas $x_2 = C/x_1$. For $x_1 < 0, x_2 > 0$, C is negative. Smaller values of C push the contour further outward. To minimize C , we move the contour as much as possible while staying in contact with the feasible region.

Step 6: Identify the optimal design

The objective is negative in the second quadrant and positive in the third quadrant, so the minimum lies on the upper boundary ($x_2 > 0$). Along the feasible boundary, the product $|x_1|x_2$ increases with x_2 on both the circular and parabolic arcs. The greatest value occurs at their intersection point in the second quadrant. Hence the optimum is

$$(x_1^*, x_2^*) = \left(\frac{1 - \sqrt{37}}{2}, \sqrt{\frac{\sqrt{37} - 1}{2}} \right) \approx (-2.541, 1.594),$$

with minimum value $f^* = x_1^* x_2^* \approx -4.05$.

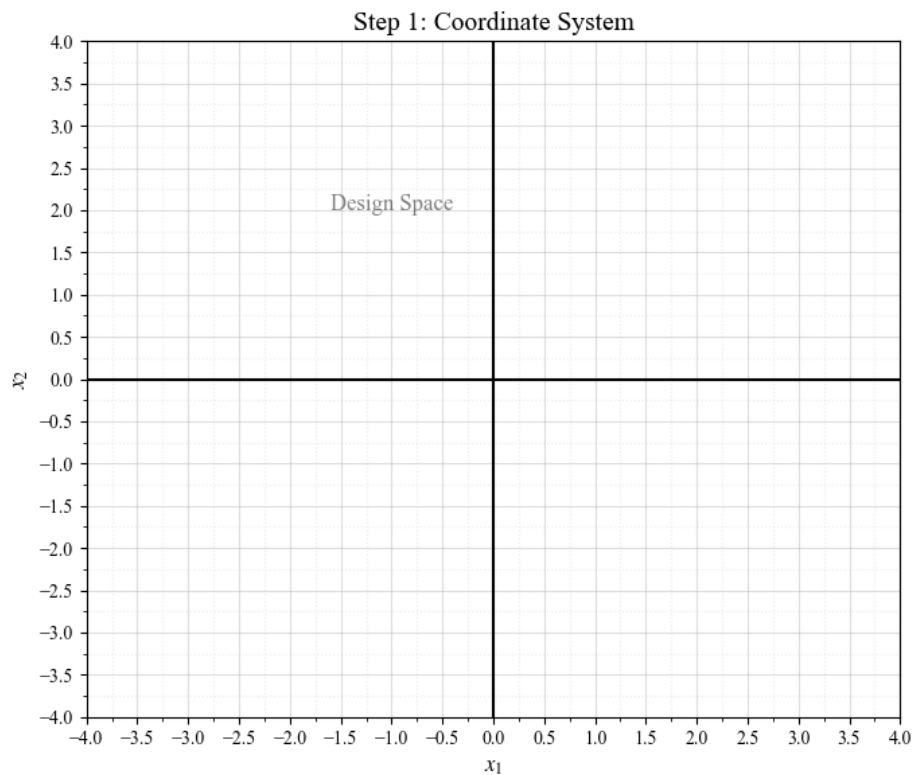


Figure 2: The suitable coordinate system

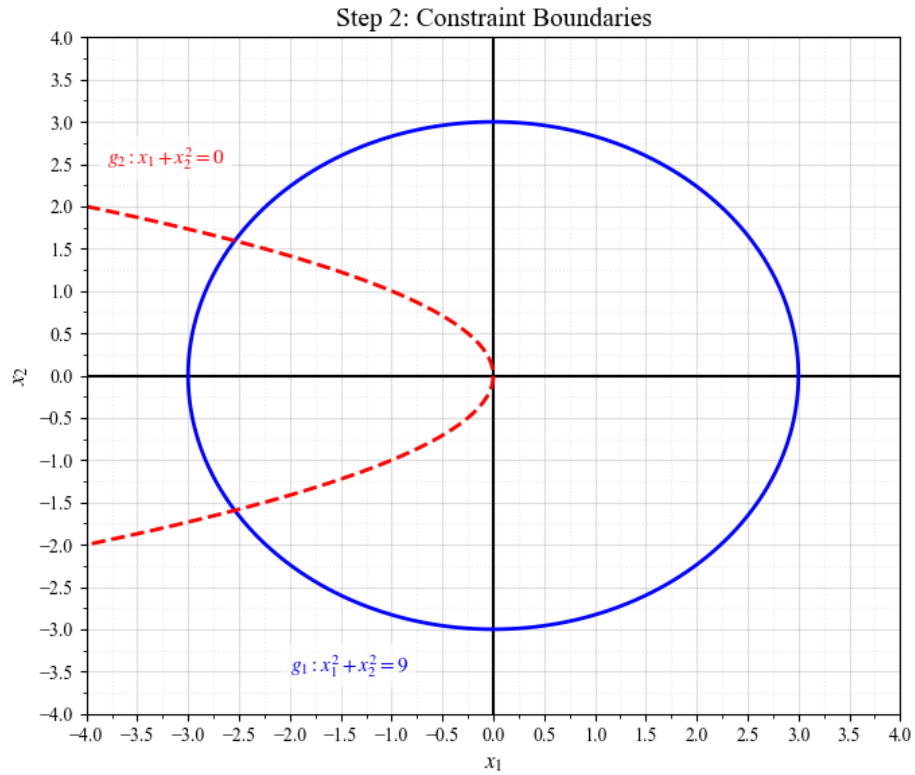


Figure 3: The constraint boundaries

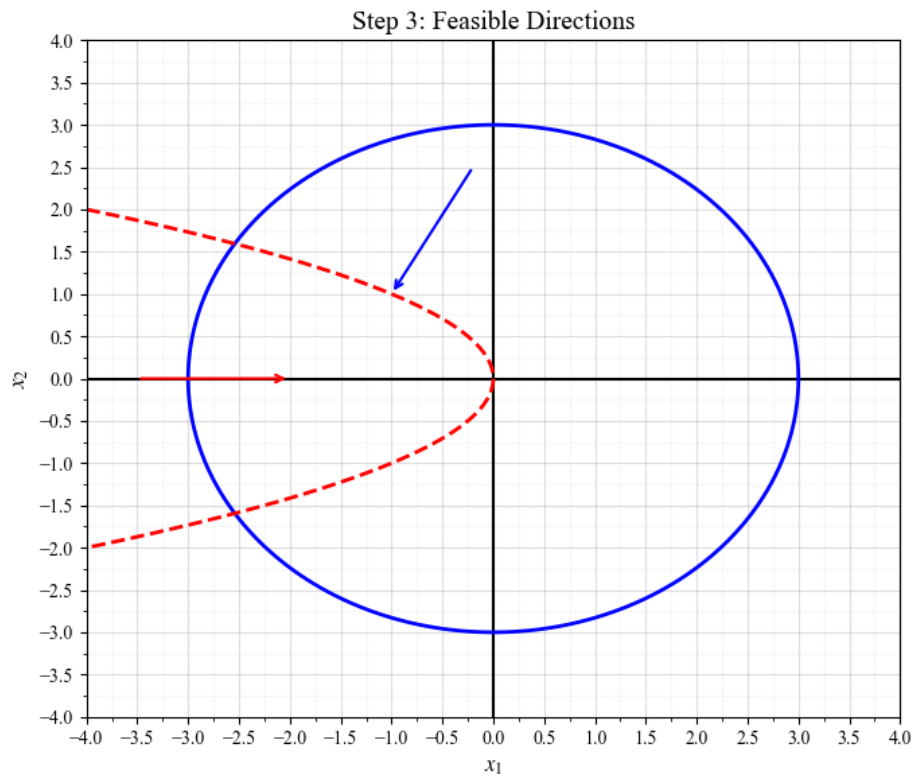


Figure 4: The feasible region for each constraint

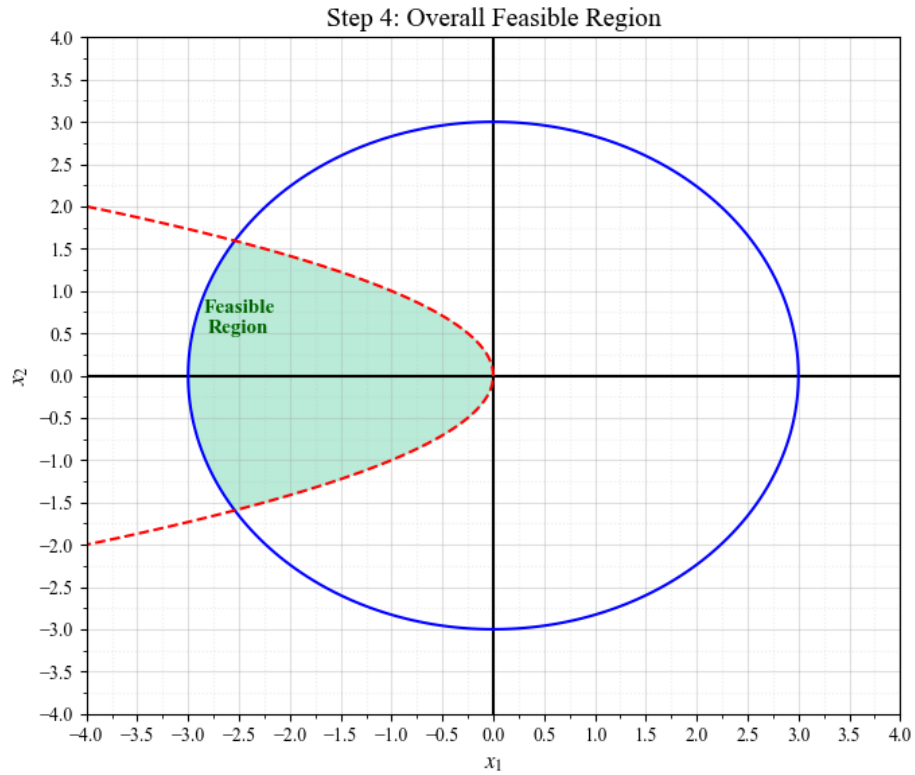


Figure 5: The overall feasible region

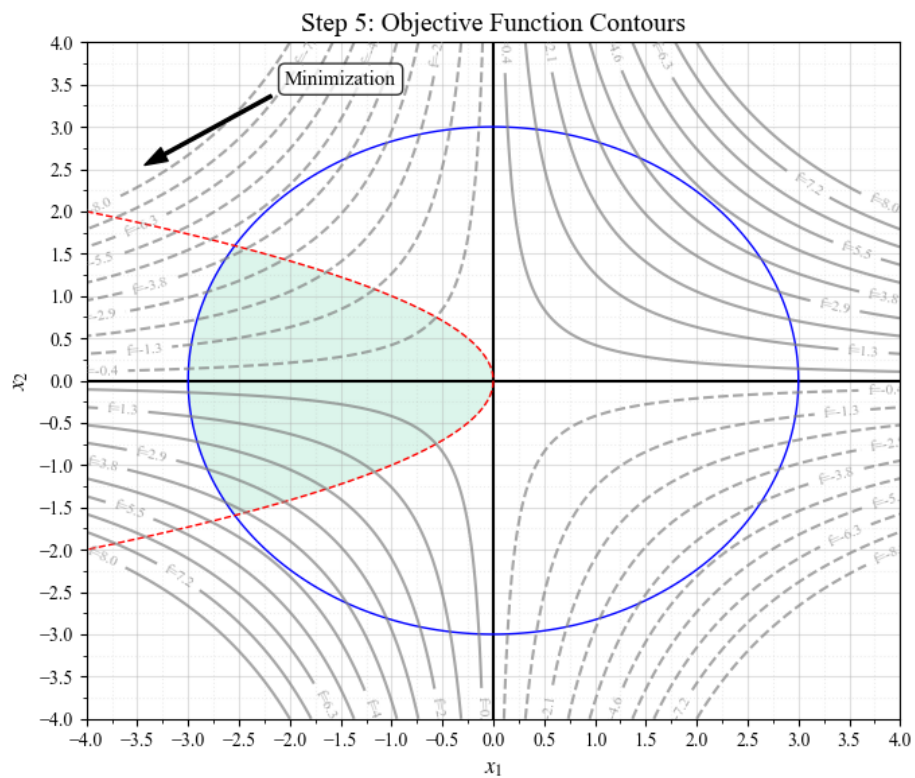


Figure 6: The objective function contours

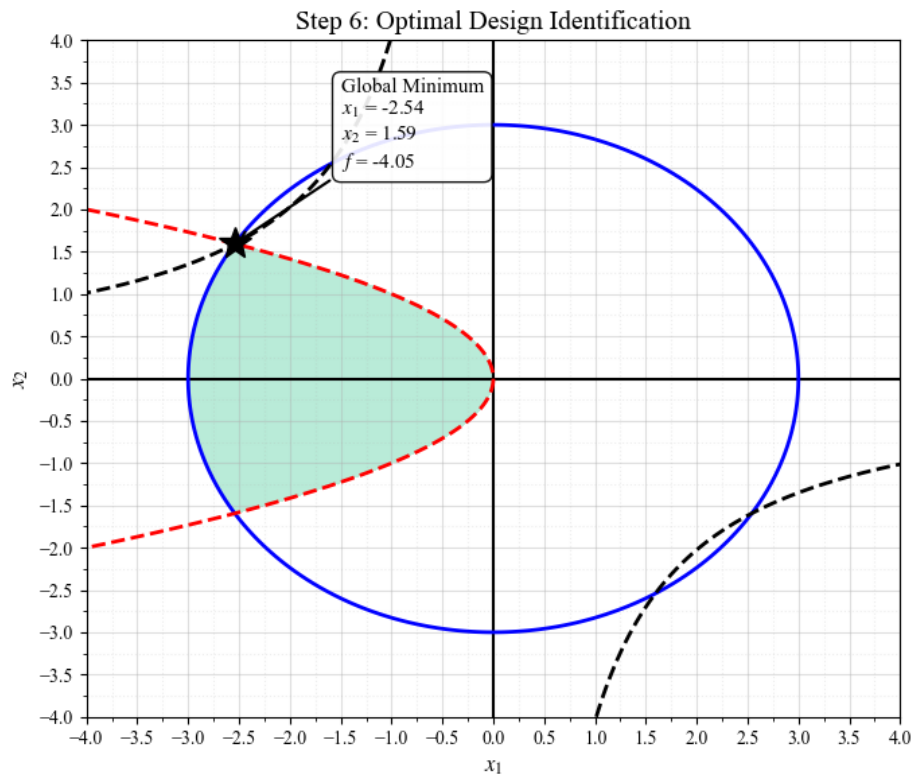


Figure 7: The optimal design

Q6: In designing a drug in tablet form composed of two additives X and Y , each tablet must provide at least 5 units of vitamin B and 4 units of vitamin C. The amounts of vitamins B and C contained in one unit of each additive, together with the cost per unit, are given in the table below.

Table 4: Vitamin content and cost per unit of additives		
	Additive X	Additive Y
Vitamin B (units/unit)	1	2
Vitamin C (units/unit)	3	2
Cost (per unit)	2	1

The amounts of additives X and Y are to be chosen so that the required minimum vitamins per tablet are supplied at the lowest cost.

- Formulate the optimal design problem.
- Solve the problem graphically.
- Solve the problem using the KKT conditions.

Solution

(a) Problem Formulation

According to the reference notes, we solve this problem in six steps.

Step 1: Clear Definition of the Problem

Design a pill combining additive X and additive Y with the lowest possible cost, ensuring the final pill contains a minimum of 5 units of Vitamin B and 4 units of Vitamin C.

Step 2: Data Collection

- Additive X : 1 unit Vit B, 3 units Vit C, Cost = 2
- Additive Y : 2 units Vit B, 2 units Vit C, Cost = 1
- Minimum Requirements: Vit B ≥ 5 , Vit C ≥ 4

Step 3: Identification and Definition of Design Variables

- x_1 : Amount of additive X in the pill.
- x_2 : Amount of additive Y in the pill.

Step 4: Identification of the Optimization Criterion

We must minimize the total cost of the pill.

$$\text{Minimize } f(x_1, x_2) = 2x_1 + x_2$$

Step 5: Identification and Modeling of Constraints

The problem requires standardizing inequalities to $g_i(X) \leq 0$:

1. Vitamin B limit:

$$x_1 + 2x_2 \geq 5 \implies g_1(x) = 5 - x_1 - 2x_2 \leq 0$$

2. Vitamin C limit:

$$3x_1 + 2x_2 \geq 4 \implies g_2(x) = 4 - 3x_1 - 2x_2 \leq 0$$

3. Non-negativity bounds (physical reality):

$$x_1 \geq 0 \implies g_3(x) = -x_1 \leq 0$$

$$x_2 \geq 0 \implies g_4(x) = -x_2 \leq 0$$

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

$$\begin{aligned} &\text{Minimize } f(X) = 2x_1 + x_2 \\ &\text{Subject to:} \\ &\quad g_1(x) = 5 - x_1 - 2x_2 \leq 0 \\ &\quad g_2(x) = 4 - 3x_1 - 2x_2 \leq 0 \\ &\quad g_3(x) = -x_1 \leq 0 \\ &\quad g_4(x) = -x_2 \leq 0 \end{aligned}$$

(b) Graphical Solution

According to the reference notes, we solve this problem in six steps.

Step 1: Create a suitable coordinate system

We establish an x_1 vs. x_2 2D plane focusing on the first quadrant (due to g_3, g_4).

Step 2: Draw the constraint boundaries

- L_1 (Vit B): $x_1 + 2x_2 = 5$. Intercepts at $(0, 2.5)$ and $(5, 0)$.
- L_2 (Vit C): $3x_1 + 2x_2 = 4$. Intercepts at $(0, 2)$ and $(1.33, 0)$.

Step 3: Identify the feasible region for each constraint

The valid area is above and to the right of both lines.

Step 4: Identify the overall feasible region

Because L_1 is strictly above L_2 everywhere in the first quadrant, g_1 dominates. The feasible region is bounded entirely by g_1 , g_3 , and g_4 . The corner points are $(0, 2.5)$ and $(5, 0)$.

Step 5: Draw the objective function contours

Contours are parallel lines of $2x_1 + x_2 = C$. They have a steeper slope (-2) than the active constraint L_1 (slope -0.5).

Step 6: Identify the optimal design

Moving the contour line towards the origin to minimize C , the last point it touches in the feasible region is $(0, 2.5)$. Minimum Cost $f(0, 2.5) = 2.5$.

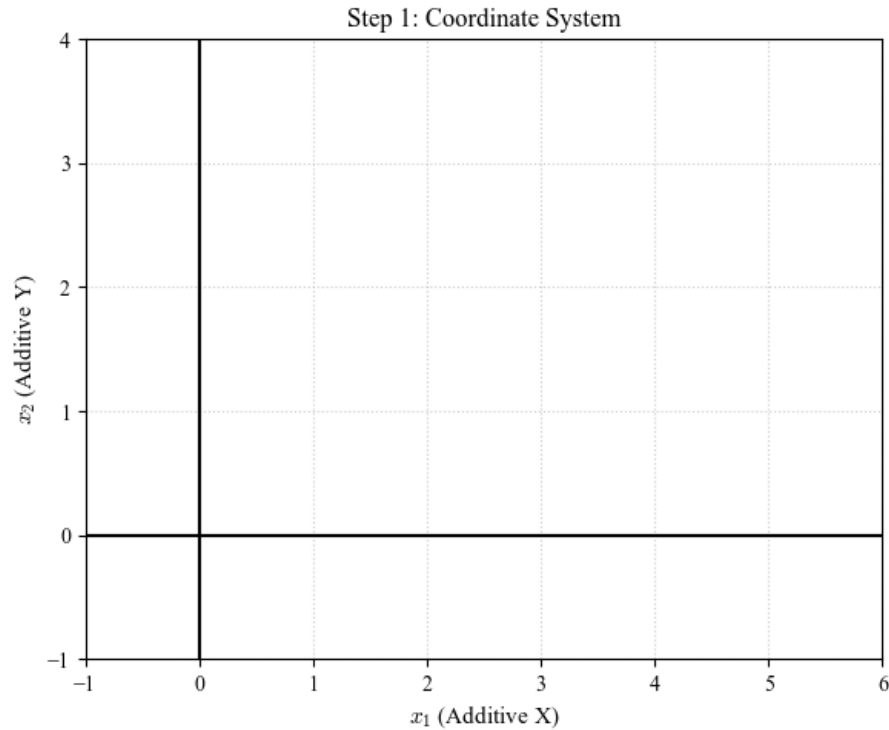


Figure 8: The suitable coordinate system

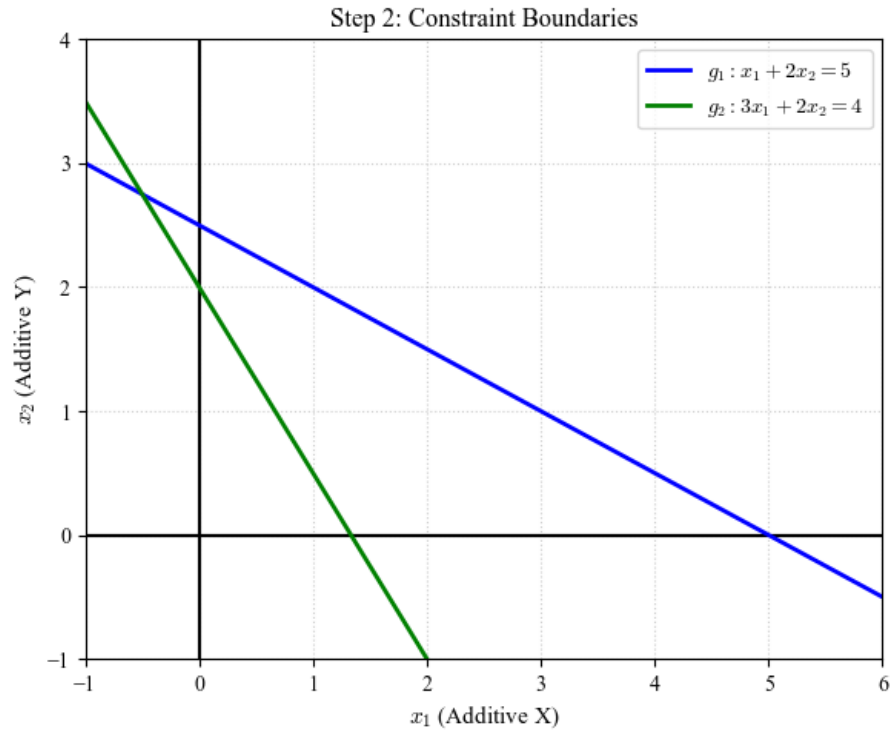


Figure 9: The constraint boundaries

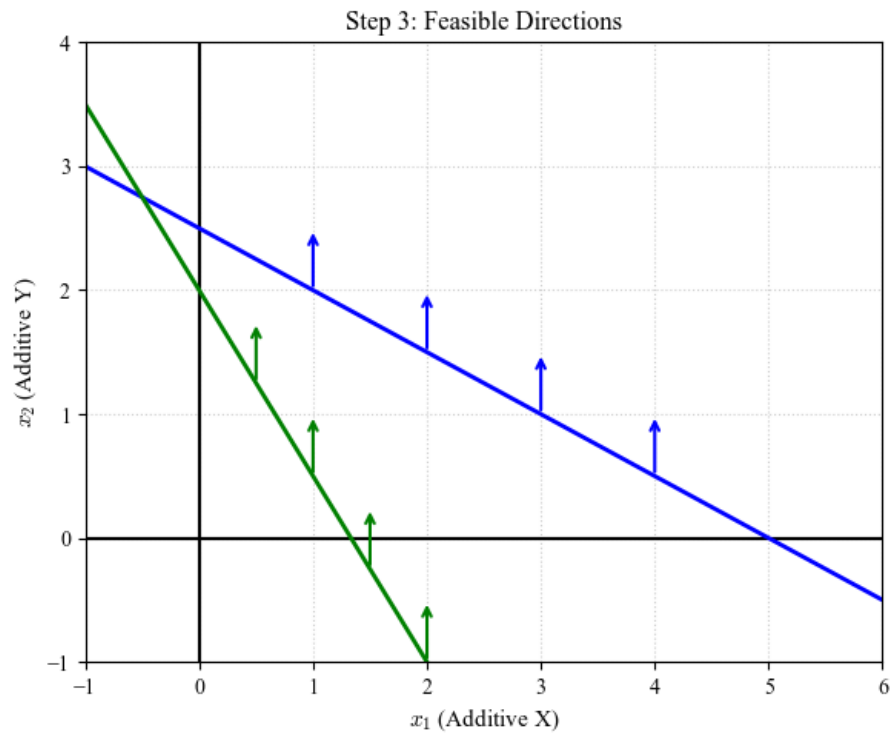


Figure 10: The feasible region for each constraint

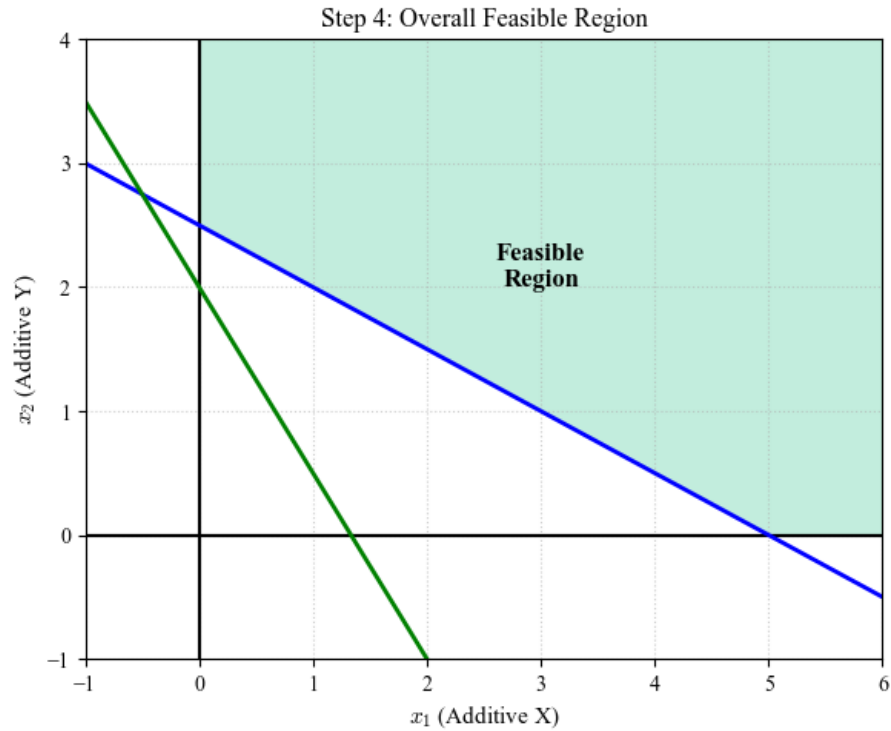


Figure 11: The overall feasible region

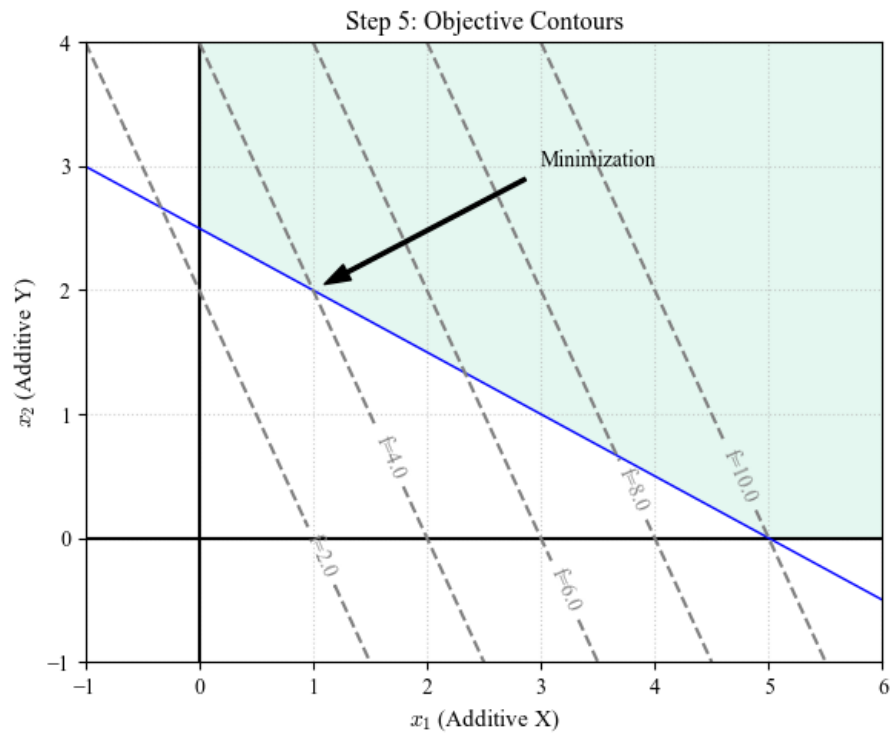


Figure 12: The objective function contours

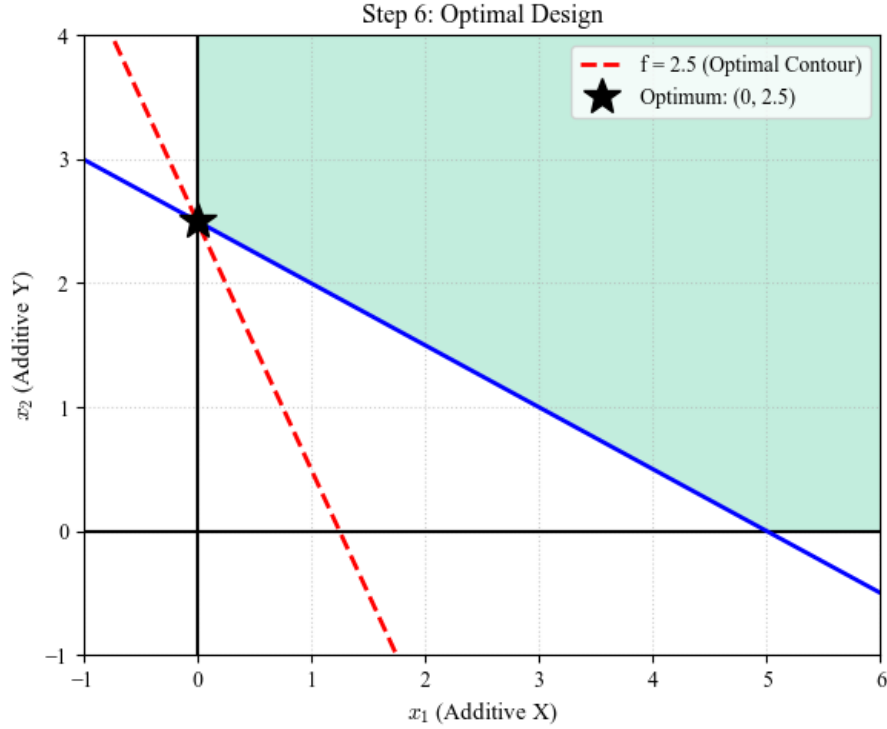


Figure 13: The optimal design

(c) KKT Method Solution

According to the KKT Conditions, the solution follows six steps.

Step 1: Form the Lagrangian Function

Introduce slack variables s_i^2 to convert inequalities to equalities

$$g_i(x) + s_i^2 = 0, \quad s_i^2 \geq 0$$

The Lagrangian with multipliers u_i (no equality multipliers v) is

$$L(x, u, s) = f(x) + \sum_{i=1}^4 u_i (g_i(x) + s_i^2)$$

$$L = 2x_1 + x_2 + u_1(5 - x_1 - 2x_2 + s_1^2) + u_2(4 - 3x_1 - 2x_2 + s_2^2) + u_3(-x_1 + s_3^2) + u_4(-x_2 + s_4^2)$$

Step 2: Gradient Conditions

$$\frac{\partial L}{\partial x_1} = 2 - u_1 - 3u_2 - u_3 = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 1 - 2u_1 - 2u_2 - u_4 = 0 \quad (2)$$

$$\frac{\partial L}{\partial u_i} = 0 \Rightarrow g_i(x) + s_i^2 = 0 \quad i = 1, \dots, 4 \quad (3a - 3d)$$

$$\frac{\partial L}{\partial s_i} = 2u_i s_i = 0 \quad i = 1, \dots, 4 \quad (4a - 4d)$$

Step 3: Check Feasible Space

From $g_i + s_i^2 = 0$ and $s_i^2 \geq 0$, we directly recover the original constraints

$$g_1 \leq 0, g_2 \leq 0, g_3 \leq 0, g_4 \leq 0$$

Step4: Check Switching Conditions

Equation (4) gives $u_i s_i = 0$ for each i . Hence,

- If $g_i < 0$ (inactive), then $s_i^2 > 0 \Rightarrow s_i \neq 0 \Rightarrow u_i = 0$.
- If $g_i = 0$ (active), then $s_i = 0$ and u_i may be ≥ 0 .

Step 5: Check Non-negativity of Lagrangian Multipliers

$$u_i \geq 0, \quad i = 1, 2, 3, 4$$

Solving the KKT system

We search for all combinations of active constraints that satisfy (1), (2) and $u_i \geq 0$. Inactive constraints have $u_i = 0$; active constraints have $g_i = 0$.

Case 1: No active constraints. Then all $u_i = 0$. Eq. (1) gives $2 = 0 \rightarrow$ impossible.

Case 2: One active constraint.

- Only g_1 active: $u_2 = u_3 = u_4 = 0$. (1) $\rightarrow u_1 = 2$; (2) $\rightarrow 1 - 4 = -3 = 0 \rightarrow$ contradiction.
- Only g_2 active: $u_2 = 2/3$; (2) $\rightarrow 1 - 4/3 \neq 0$.
- Only g_3 or g_4 active: lead to $2 = 0$ or $1 = 0$ in (1)/(2). No solution.

Case 3: Two active constraints. We test all pairs.

- Active g_1, g_2 : $u_3 = u_4 = 0$. Solve (1) $2 - u_1 - 3u_2 = 0$ and (2) $1 - 2u_1 - 2u_2 = 0$.

Solution: $u_2 = 0.75, u_1 = -0.25 \rightarrow$ violates $u_1 \geq 0$.

- Active g_1, g_3 : $u_2 = u_4 = 0$. (2) $\rightarrow 1 - 2u_1 = 0 \Rightarrow u_1 = 0.5$ and (1) $\rightarrow 2 - u_1 - u_3 = 0 \Rightarrow u_3 = 1.5$.

All $u_i \geq 0$. Feasibility: $g_1 = 0 \Rightarrow x_1 + 2x_2 = 5, g_3 = 0 \Rightarrow x_1 = 0 \rightarrow x_2 = 2.5$.

Check other constraints: $g_2 = 4 - 0 - 5 = -1 \leq 0; g_4 = -2.5 \leq 0$. Feasible.

Candidate point: $(x_1, x_2) = (0, 2.5)$ with $f = 2.5$.

- Active g_1, g_4 : $u_2 = u_3 = 0$. (1) $\rightarrow u_1 = 2$; (2) $\rightarrow 1 - 4 - u_4 = 0 \Rightarrow u_4 = -3 \rightarrow$ violates $u_4 \geq 0$.

- Active g_2, g_3 : $g_2 = 0, g_3 = 0 \Rightarrow x_1 = 0, x_2 = 2$. Check $g_1 = 5 - 0 - 4 = 1 > 0 \rightarrow$ infeasible.
- Active g_2, g_4 : $g_2 = 0, g_4 = 0 \Rightarrow x_2 = 0, x_1 = 4/3$. $g_1 = 5 - 4/3 > 0 \rightarrow$ infeasible.
- Active g_3, g_4 : $x_1 = x_2 = 0, g_1 = 5 > 0 \rightarrow$ infeasible.

Case 4: Three or four active constraints. No solution satisfies all equalities simultaneously. The unique KKT point is $x^* = (0, 2.5)$ with multipliers $u_1^* = 0.5, u_3^* = 1.5, u_2^* = u_4^* = 0$.

Step 6: Check Regularity of Candidate Points

At $x^* = (0, 2.5)$, the active constraints are g_1 and g_3 . Their gradients are

$$\nabla g_1 = (-1, -2), \quad \nabla g_3 = (-1, 0)$$

These vectors are linearly independent, so x^* is a regular point and the KKT conditions are applicable.

Optimal design

Use 0 units of additive X and 2.5 units of additive Y , giving a minimum cost of $f^* = 2.5$.

$x_1^* = 0, \quad x_2^* = 2.5, \quad f^* = 2.5$

Q7: Suppose a company has invited you as a consultant to design a flagpole of height H . You are asked to design the pole to have minimum mass. The pole will be made of a uniform hollow circular tube with outer diameter d_o and inner diameter d_i . The pole is located in an area exposed to strong wind currents. Based on investigations, the wind force has two components: a distributed load and a concentrated force. That is, the wind load on the flag is modeled as a uniformly distributed load of intensity w (kN/m) along the pole and a concentrated force P (kN) applied at the tip of the flagpole.

The flagpole must not fail due to bending or shear under the applied forces. Furthermore, the maximum tip deflection must not exceed 10 cm. The ratio of mean diameter to thickness must not exceed 60.

The bounds on the design variables are: $5 \leq d_o \leq 50$ cm and $4 \leq d_i \leq 45$ cm. Additional data are provided in Table 5. If you need any other information, you may make assumptions and proceed with the calculations, stating your assumptions clearly.

The requirements of the flagpole optimal design problem are:

1. Formulate the optimal design problem.
2. Solve using graphical optimization.
3. Solve using the KKT conditions.
4. If the distributed load w and the concentrated load P increase to 2.1 kN/m and 4.1 kN, respectively, what will be the effect of these changes on the mass of the flagpole?

Table 5: Data and formulas for the flagpole design

Item	Data/Formula
Cross-sectional area	$A = \frac{\pi}{4}(d_o^2 - d_i^2)$
Moment of inertia	$I = \frac{\pi}{64}(d_o^4 - d_i^4)$
Modulus of elasticity	$E = 210$ GPa
Allowable bending stress	$\sigma_b = 165$ MPa
Allowable shear stress	$\tau_s = 50$ MPa
Mass density of pole material	$\rho = 7800$ kg/m ³
Wind load	$w = 2.0$ kN/m
Height of flag pole	$H = 10$ m
Concentrated load at top	$P = 4.0$ kN
Moment at base	$M = PH + 0.5 wH^2$ (kN·m)
Bending stress	$\sigma = \frac{M}{2I}$ (kPa)
Shear at base	$S = P + wH$ (kN)
Shear stress	$\tau = \frac{S}{12I}(d_o^2 + d_i d_o + d_i^2)$ (kPa)
Deflection at top	$\delta = \frac{PH^3}{3EI} + \frac{wH^4}{8EI}$
Minimum and maximum thickness	0.5 cm and 2 cm

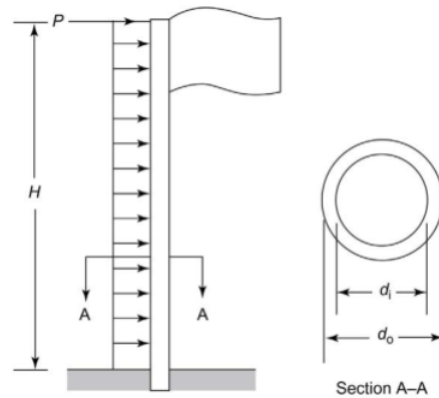


Figure 14: A schematic of the flagpole design

Solution

(a) Problem Formulation

According to the reference notes, we solve this problem in five steps.

Step 1: Clear Definition of the Problem

Design a hollow circular flagpole to minimize its total mass while withstanding specific wind and concentrated loads. The design must not fail due to bending or shear, must restrict deflection at the top to 10 cm, and must adhere to specific manufacturing thickness and ratio limits.

Step 2: Data Collection

- $E = 210 \text{ GPa}$, $\rho = 7800 \text{ kg/m}^3$
- $H = 10 \text{ m}$
- $P = 4.0 \text{ kN}$, $w = 2.0 \text{ kN/m}$
- Moment at base: $M = PH + 0.5wH^2 = 140 \text{ kN}\cdot\text{m}$
- Max Deflection $\delta \leq 0.1 \text{ m}$
- Max Bending Stress $\sigma_b \leq 165 \text{ MPa}$
- Diameter-to-thickness ratio limit ≤ 60
- Thickness limit: $0.5 \text{ cm} \leq t \leq 2.0 \text{ cm}$

Step 3: Identification and Definition of Design Variables

- d_o : Outer diameter of the flagpole (cm)
- d_i : Inner diameter of the flagpole (cm)

Step 4: Identification of the Optimization Criterion

Since height and density are constant, minimizing the mass is identical to minimizing the cross-sectional area A .

$$\text{Minimize } f(d_o, d_i) = \frac{\pi}{4}(d_o^2 - d_i^2)$$

Step 5: Identification and Modeling of Constraints

Converting all structural formulas to standard limits in cm:

1. Deflection Limit: $\delta = \frac{PH^3}{3EI} + \frac{wH^4}{8EI} \leq 0.1 \implies I \geq 18,254 \text{ cm}^4$. Using $I = \frac{\pi}{64}(d_o^4 - d_i^4)$, this becomes:

$$g_1(d_o, d_i) : 371870 - d_o^4 + d_i^4 \leq 0$$

2. Bending Stress: $\sigma = \frac{M(d_o/2)}{I} \leq 165 \text{ MPa} \implies I \geq 424.24d_o$.

$$g_2(d_o, d_i) : 8642.5d_o - d_o^4 + d_i^4 \leq 0$$

3. Ratio Limit: $\frac{d_{avg}}{t} = \frac{d_o + d_i}{d_o - d_i} \leq 60 \implies 61d_i \leq 59d_o$.

$$g_3(d_o, d_i) : d_i - \frac{59}{61}d_o \leq 0$$

4. Thickness Limits: $t = \frac{d_o - d_i}{2}$ must be between 0.5 and 2 cm.

$$g_4(d_o, d_i) : 1 - d_o + d_i \leq 0$$

$$g_5(d_o, d_i) : d_o - d_i - 4 \leq 0$$

5. Physical Bounds: $5 \leq d_o \leq 50$ and $4 \leq d_i \leq 45$.

$$g_6(d_o, d_i) : 5 - d_o \leq 0$$

$$g_7(d_o, d_i) : d_o - 50 \leq 0$$

$$g_8(d_o, d_i) : 4 - d_i \leq 0$$

$$g_9(d_o, d_i) : d_i - 45 \leq 0$$

6. Shear stress limit: $\tau = \frac{S}{12I}(d_o^2 + d_i d_o + d_i^2) \leq 50 \text{ MPa}$ where $S = P + wH = 24,000 \text{ N}$ and $I = \frac{\pi}{64}(d_o^4 - d_i^4)$ is the moment of inertia. The inequality constraint is

$$g_{10}(d_o, d_i) : \frac{25.6}{\pi} \frac{d_o^2 + d_i d_o + d_i^2}{d_o^4 - d_i^4} - 1 \leq 0$$

Final Standard Formulation

Putting it all together into the standard constrained optimization format, Finally we have

$$\begin{aligned}
 &\text{Minimize } f(d_o, d_i) = \frac{\pi}{4}(d_o^2 - d_i^2) \\
 &\text{Subject to:} \\
 &\quad g_1(d_o, d_i) = 371870 - d_o^4 + d_i^4 \leq 0 \\
 &\quad g_2(d_o, d_i) = 8642.5d_o - d_o^4 + d_i^4 \leq 0 \\
 &\quad g_3(d_o, d_i) = d_i - (59/61)d_o \leq 0 \\
 &\quad g_4(d_o, d_i) = 1 - d_o + d_i \leq 0 \\
 &\quad g_5(d_o, d_i) = d_o - d_i - 4 \leq 0 \\
 &\quad g_6(d_o, d_i) = 5 - d_o \leq 0 \\
 &\quad g_7(d_o, d_i) = d_o - 50 \leq 0 \\
 &\quad g_8(d_o, d_i) = 4 - d_i \leq 0 \\
 &\quad g_9(d_o, d_i) = d_i - 45 \leq 0 \\
 &\quad g_{10}(d_o, d_i) = \frac{25.6}{\pi} \frac{d_o^2 + d_i d_o + d_i^2}{d_o^4 - d_i^4} - 1 \leq 0
 \end{aligned}$$

(b) Graphical Solution

According to the reference notes, we solve this problem in six steps.

Step 1: Create a suitable coordinate system

We plot d_o on the X-axis (35 to 50 cm) and d_i on the Y-axis (34 to 48 cm).

Step 2: Draw the constraint boundaries

Plot $d_o^4 - d_i^4 = 371870$, $d_i = \frac{59}{61}d_o$, $d_i = d_o - 1$, and $d_i = d_o - 4$.

Step 3: Identify the feasible region for each constraint

We must be above the deflection limit curve, below the ratio line, and between the thickness lines.

Step 4: Identify the overall feasible region

The bounded region is shaped like a sliver, squeezed between the Ratio limit (top edge) and the Deflection limit (bottom edge).

Step 5: Draw the objective function contours

The contours of $Area = C$ form curves. To minimize area, we want d_i to be as close to d_o as possible (thinnest wall). The direction of decrease points toward the upper-left of the feasible region.

Step 6: Identify the optimal design

The lowest possible area within the feasible region occurs exactly at the intersection of the Deflection Limit (g_1) and the Ratio Limit (g_3). Solving $d_i = 0.9672d_o$ and $d_o^4 - d_i^4 = 371870$ simultaneously yields the optimal point: $d_o = 41.56$ cm, $d_i = 40.20$ cm

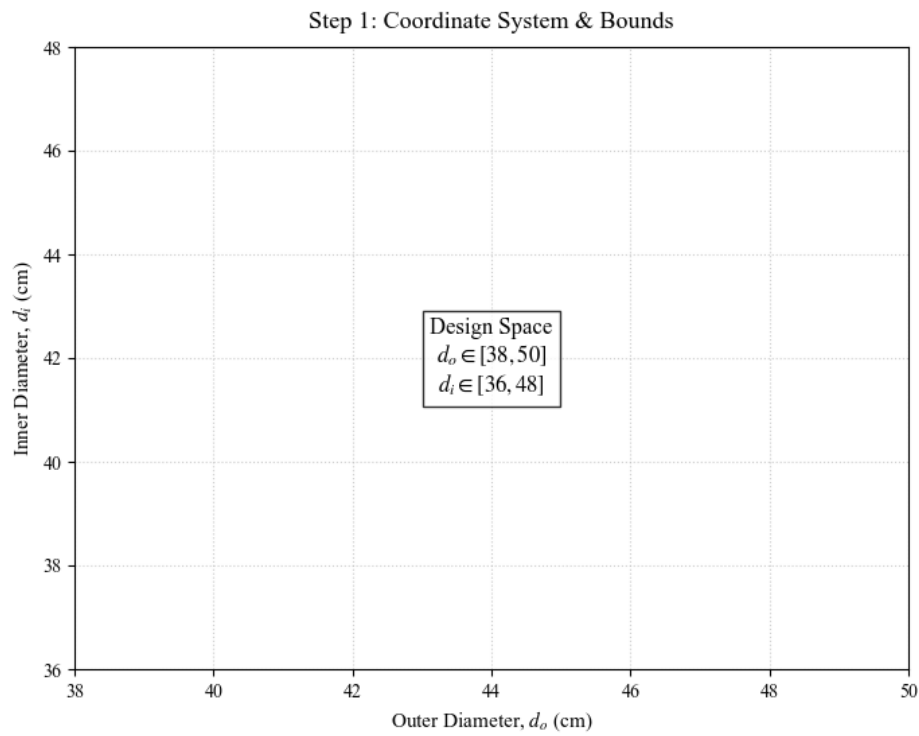


Figure 15: The suitable coordinate system

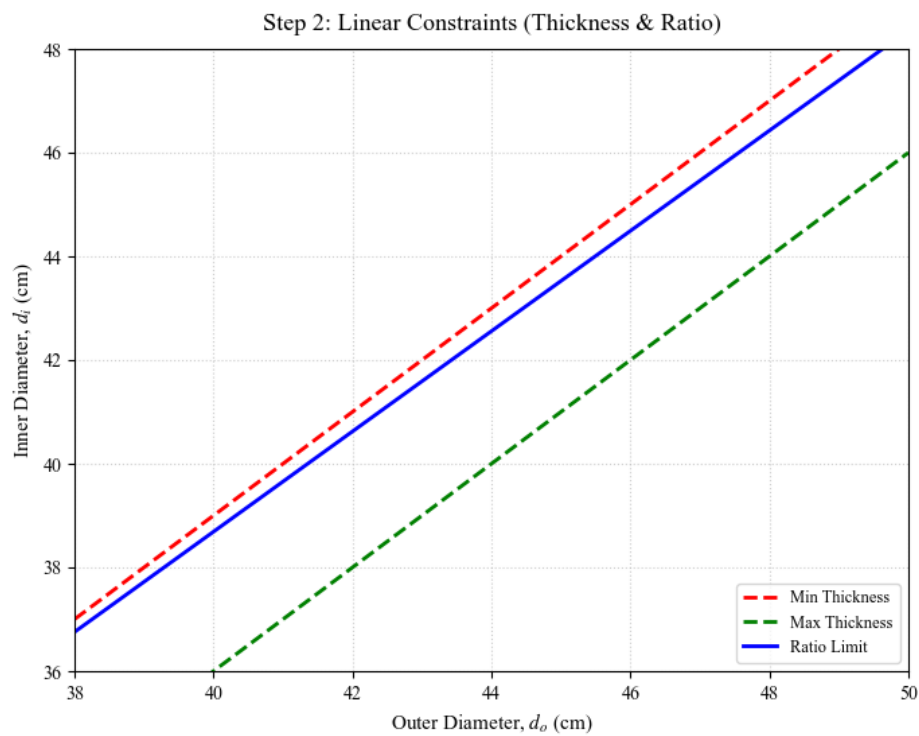


Figure 16: The constraint boundaries

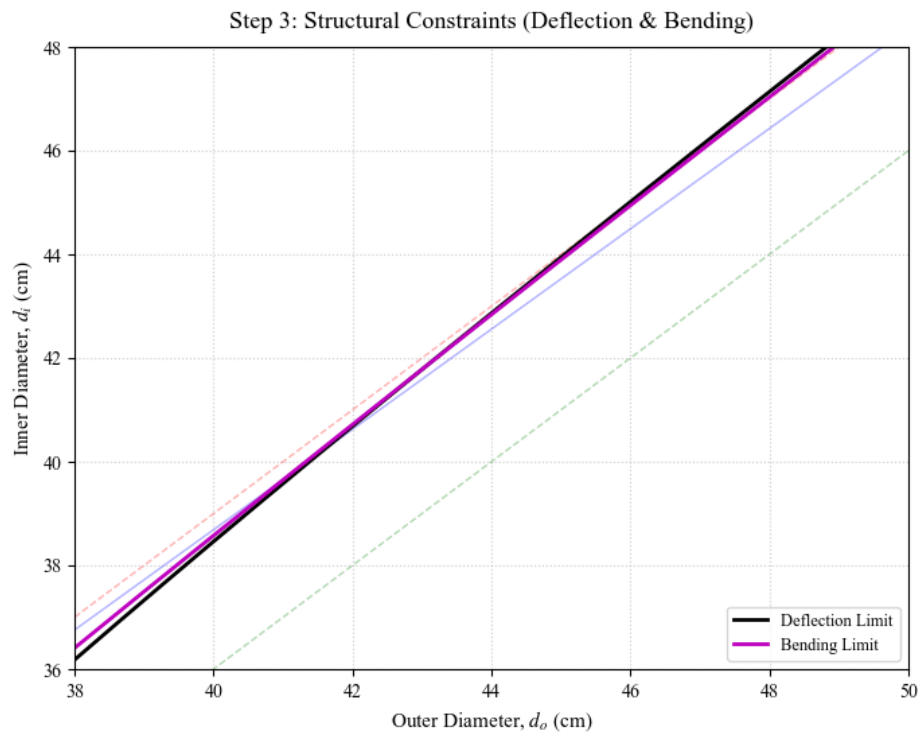


Figure 17: The feasible region for each constraint

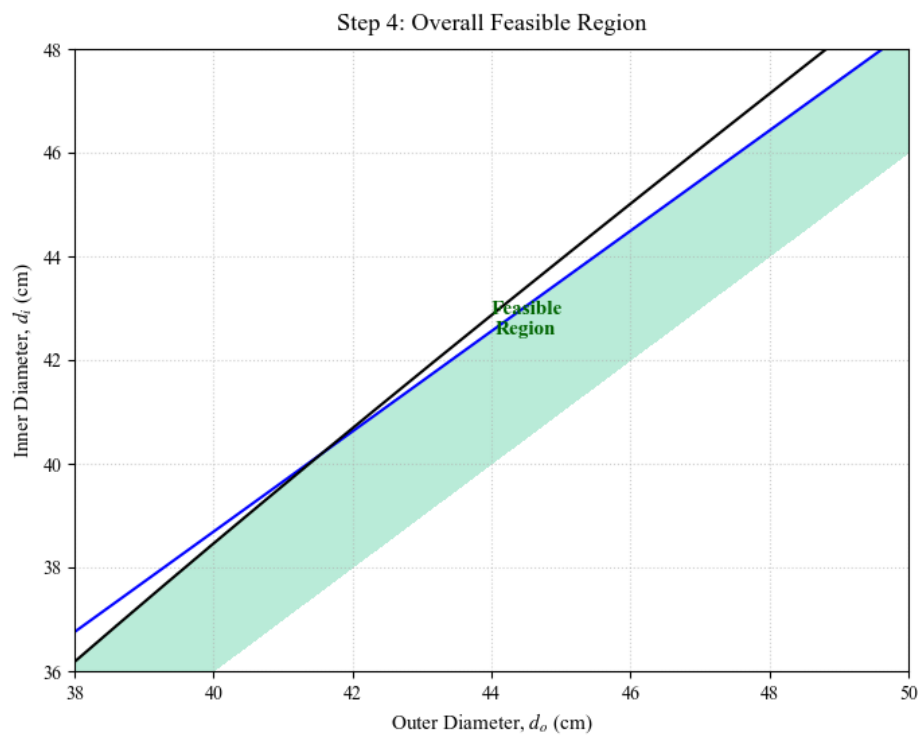


Figure 18: The overall feasible region

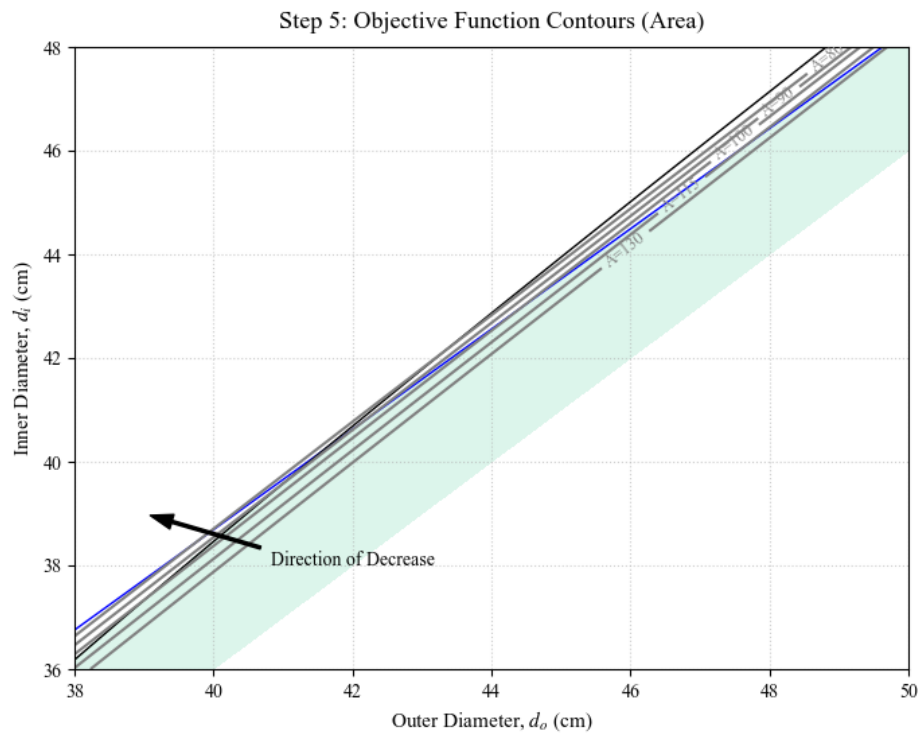


Figure 19: The objective function contours

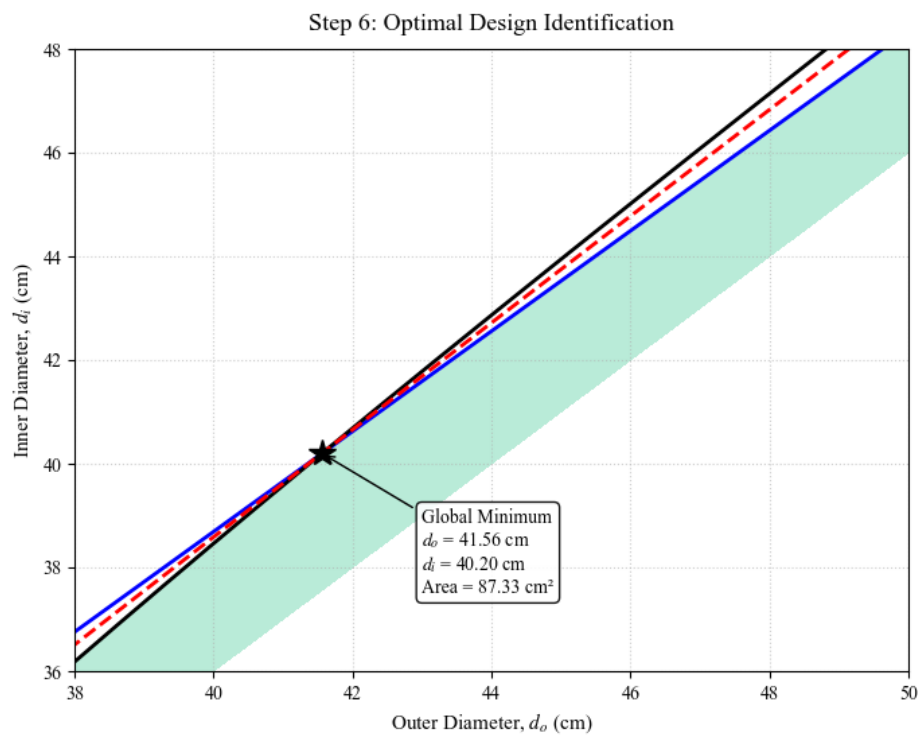


Figure 20: The optimal design

(c) KKT Method Solution

According to the KKT Conditions, the solution follows six steps.

Step 1: Form the Lagrangian Function

$$L = f(d_o, d_i) + \sum_{j=1}^{10} u_j (g_j(d_o, d_i) + s_j^2)$$

where u_j are Lagrange multipliers and s_j are real slack variables.

Step 2: Gradient Conditions

$$\frac{\partial L}{\partial d_o} = 0, \quad \frac{\partial L}{\partial d_i} = 0, \quad \frac{\partial L}{\partial u_j} = 0 \implies g_j + s_j^2 = 0.$$

Note: Substituting after simplify coefficients.

Step 3: Check Feasible Space

Candidates must satisfy $g_j(x, y) \leq 0$ for all j

Step 4: Check Switching Conditions

$$\frac{\partial L}{\partial s_j} = 0 \implies 2u_j s_j = 0$$

Together with $g_j + s_j^2 = 0$ this gives the complementary slackness conditions $u_j g_j = 0$.

Step 5: Check Non-negativity of Lagrangian Multipliers

$u_j \geq 0$ for all j .

Because the objective decreases when d_o is made smaller and d_i is made larger, the optimum is expected to lie on the geometric boundary that limits the diameter ratio (g_3) and on the deflection constraint (g_1). All other constraints turn out to be inactive. Therefore we set

$$u_2 = u_4 = u_5 = u_6 = u_7 = u_8 = 0$$

Computing the partial derivatives:

$$\frac{\partial L}{\partial d_o} = 2d_o - 4d_o^3 u_1 - \frac{59}{61} u_3 = 0 \quad (5)$$

$$\frac{\partial L}{\partial d_i} = -2d_i + 4d_i^3 u_1 + u_3 = 0 \quad (6)$$

and require $g_1 = 0$ and $g_3 = 0$.

$$g_3 = 0 \implies d_i = \frac{59}{61} d_o \quad (7)$$

$$g_1 = 0 \implies d_o^4 - d_i^4 = 371870 \quad (8)$$

Substitute (7) into (8),

$$d_o^4 - \left(\frac{59}{61}\right)^4 d_o^4 = 371870 \implies \left[1 - \left(\frac{59}{61}\right)^4\right] d_o^4 = 371870.$$

$$d_o = (2978832.65)^{1/4} \approx 41.544 \text{ cm}$$

$$d_i = \frac{59}{61} \times 41.544 \approx 40.181 \text{ cm}$$

Inactive constraints:

$$\begin{aligned} g_2(d_o, d_i) &= 8642.5d_o - d_o^4 + d_i^4 = -12195.92 \leq 0 \\ g_4(d_o, d_i) &= 1 - d_o + d_i = -0.36 \leq 0 \\ g_5(d_o, d_i) &= d_o - d_i - 4 = -2.64 \leq 0 \\ g_6(d_o, d_i) &= 5 - d_o = -36.54 \leq 0 \\ g_7(d_o, d_i) &= d_o - 50 = -8.46 \leq 0 \\ g_8(d_o, d_i) &= 4 - d_i = -36.180 \\ g_9(d_o, d_i) &= d_i - 45 = -4.82 \leq 0 \\ g_{10}(d_o, d_i) &= \frac{25.6}{\pi} \frac{d_o^2 + d_i d_o + d_i^2}{d_o^4 - d_i^4} - 1 = -0.872 \leq 0 \end{aligned}$$

Now compute the active multipliers u_1 and u_3 from (5) and (6) with all other $u_j = 0$

$$(5) \quad 2d_o - 4d_o^3 u_1 - \frac{59}{61} u_3 = 0$$

$$(6) \quad -2d_i + 4d_i^3 u_1 + u_3 = 0$$

From (6): $u_3 = 2d_i - 4d_i^3 u_1$

Substitute into (5):

$$2d_o - 4d_o^3 u_1 - \frac{59}{61} (2d_i - 4d_i^3 u_1) = 0$$

$$2d_o - \frac{118}{61} d_i - u_1 (4d_o^3 - \frac{236}{61} d_i^3) = 0$$

Using $d_i = \frac{59}{61} d_o$:

$$2d_o - \frac{118}{61} \cdot \frac{59}{61} d_o = d_o \left(2 - \frac{6962}{3721}\right) = \frac{480}{3721} d_o > 0$$

$$4d_o^3 - \frac{236}{61} \left(\frac{59}{61}\right)^3 d_o^3 = d_o^3 \left[4 - \frac{236 \cdot 59^3}{61^4}\right] = 0.5d_o^3$$

$$u_1 = \frac{\frac{480}{3721} d_o}{0.5d_o^3} = \frac{960}{3721d_o^2} \approx \frac{0.258}{d_o^2}$$

With $d_o = 41.544$,

$$u_1 \approx \frac{0.258}{1725.6} \approx 1.50 \times 10^{-4} \geq 0$$

$$u_3 = 2d_i - 4d_i^3 u_1 = 2(40.181) - 4(40.181^3)(1.50 \times 10^{-4}) = 41.52 \geq 0$$

Both multipliers are non-negative, satisfying Step 5.

Step 6: Check Regularity of Candidate Points

$$\nabla g_1 = (-4d_o^3, 4d_i^3) \approx (-286600, 259500), \quad \nabla g_3 = \left(-\frac{59}{61}, 1\right) \approx (-0.967, 1)$$

These vectors are clearly not parallel, hence the point is regular (Step 6 satisfied).

Optimal design

$$d_o^* \approx 41.54 \text{ cm}, \quad d_i^* \approx 40.18 \text{ cm}$$

Note: The difference from the result obtained from the graphical solution is due to a calculation error.

(d) Post-Optimality

Step 1: Updated expression of the deflection constraint

The maximum deflection is

$$\delta = \frac{PH^3}{3EI} + \frac{wH^4}{8EI} \leq 0.1 \text{ m}$$

With $H = 10 \text{ m}$, $E = 210 \text{ GPa}$, I in m^4 , and converting to cm^4

$$I_{\min}^{\text{new}} (\text{cm}^4) = \frac{100}{63}P + \frac{100}{16.8}w = 1.58730P + 5.95238w$$

Using $P = 4100 \text{ N}$, $w = 2100 \text{ N/m}$:

$$I_{\min}^{\text{new}} = 1.58730(4100) + 5.95238(2100) = 19007.93 \text{ cm}^4$$

The constraint g_1 in standard form is

$$g_1(d_o, d_i) = \frac{64}{\pi} I_{\min}^{\text{new}} - d_o^4 + d_i^4 \leq 0$$

so the new constant is

$$C_1^{\text{new}} = \frac{64}{\pi} \times 19007.93 = 387230$$

Thus

$$g_1(d_o, d_i) : 387230 - d_o^4 + d_i^4 \leq 0$$

Step 2: Updated bending stress constraint

The moment at the base is

$$M = PH + \frac{1}{2}wH^2 = 4.1(10) + 0.5(2.1)(100) = 41 + 105 = 146 \text{ kN}\cdot\text{m}$$

The bending stress constraint g_2 becomes

$$g_2(d_o, d_i) = \frac{32M}{\pi\sigma_b}d_o - d_o^4 + d_i^4 \leq 0$$

with $\sigma_b = 165 \text{ MPa} = 16500 \text{ N/cm}^2$, $M = 1.46 \times 10^7 \text{ N}\cdot\text{cm}$

$$\frac{32M}{\pi\sigma_b} = \frac{32(1.46 \times 10^7)}{\pi(16500)} \approx 9015$$

Hence

$$g_2(d_o, d_i) : 9015d_o - d_o^4 + d_i^4 \leq 0$$

All other constraints (g_3 to g_8) remain unchanged and inactive.

Step 3: KKT solution with the new constants

As argued in part (c), the mass-minimising design is driven by the deflection limit (g_1) and the diameter-ratio limit (g_3).

$$\begin{aligned} g_3 = 0 : y &= \frac{59}{61}x \\ g_1 = 0 : d_o^4 - d_i^4 &= C_1^{\text{new}} = 387\,230 \end{aligned}$$

Substituting y :

$$\begin{aligned} d_o^4 - \left(\frac{59}{61}\right)^4 d_o^4 &= 387\,230 \implies \left[1 - \left(\frac{59}{61}\right)^4\right] d_o^4 = 387\,230 \rightarrow d_o \approx 41.97 \text{ cm} \\ d_i &= \frac{59}{61}(41.97) \approx 40.59 \text{ cm} \end{aligned}$$

$$\boxed{d_o \approx 41.97 \text{ cm}, \quad d_i \approx 40.59 \text{ cm}}$$

Inactive unchanged constraints:

$$\begin{aligned} g_2(d_o, d_i) &= 8642.5d_o - d_o^4 + d_i^4 = -12195.92 \leq 0 \\ g_4(d_o, d_i) &= 1 - d_o + d_i = -0.38 \leq 0 \\ g_5(d_o, d_i) &= d_o - d_i - 4 = -2.62 \leq 0 \\ g_6(d_o, d_i) &= 5 - d_o = -36.97 \leq 0 \\ g_7(d_o, d_i) &= d_o - 50 = -8.03 \leq 0 \\ g_8(d_o, d_i) &= 4 - d_i = -36.59 \leq 0 \\ g_9(d_o, d_i) &= d_i - 45 = -4.41 \leq 0 \\ g_{10}(d_o, d_i) &= \frac{25.6 d_o^2 + d_i d_o + d_i^2}{d_o^4 - d_i^4} - 1 = -0.8753 \leq 0 \end{aligned}$$

The previous minimum mass of the flagpole is

$$m = \rho A H = 7800 \text{ kg/m}^3 \times \frac{\pi}{40000} (41.54^2 - 40.18^2) \text{ m}^2 \times 10 \text{ m} \approx 681.22 \text{ kg}$$

And the new minimum mass of the flagpole is

$$m' = \rho A' H = 7800 \text{ kg/m}^3 \times \frac{\pi}{40000} (41.97^2 - 40.59^2) \text{ m}^2 \times 10 \text{ m} \approx 697.22 \text{ kg}$$

So,

$$\boxed{m' - m = 697.22 - 681.22 = 16 \text{ kg}}$$

The mass of the flagpole must increase by 16 kg to handle the new loads.